

Small Assignment: Nested Structure and Simple Class

Write a C++ program to model a person's detail with address information using structures and a class.

1. Define a nested structure

Create a structure named `Address` containing:

- `houseNo` (integer)
- `street` (string)
- `city` (string)
- `state` (string)
- *Inside* `Address`, define another structure called `PIN` with:
 - `code` (integer)
 - `country` (string)

2. Define a simple class

Create a class `Person` with **private** data members:

- `name` (string)
- `age` (integer)
- `addr` (an object of type `Address`)

Provide **public** member functions:

- A parameterised constructor to initialise `name`, `age`, and all fields of `addr` (including nested `PIN`).
- `void display()` — prints all person details including full address and `PIN`.

3. Demonstrate in `main()`

- Create a `Person` object with sample data.
- Call `display()` to show the output.

Expected output format (example):

Name: Alice

Age: 30

Address: 42, Park Lane, Springfield, Stateland

PIN: 123456, USA

This assignment covers:

- **Structure** (`Address`)
- **Nested structure** (`PIN` inside `Address`)
- **Simple class** (`Person`) with encapsulation and a constructor.